A little guidance

The light we cannot see. · Grade 2 · Parent answer key

Three gentle sessions: 15 minutes each. Understanding matters more than finishing every line.

Math · 1

6 visible-color places: 7 − 1 = 6.

Math · 2

6 hops from place 1 to place 7. Count the gaps, not the starting place.

Math · 3

3 places farther along: 7 − 4 = 3. Extension: the UV-A end has more closely spaced crests.

Reading · 1

Light-sensitive material darkened beyond violet, showing that unseen light reached it.

Reading · 2

The coating absorbs UV-A and emits visible light. The star has no tiny lamps inside.

Writing

Accept a labeled drawing, dictation, or 1–3 sentences that connect a clue to the light path.

Let a guess come first. This is a screen and paper exploration; no real lamp or chemical experiment is needed. Ordinary black lights mainly use near-ultraviolet UV-A, not X-rays. Adults manage real UV equipment; never aim it at eyes or skin. UV-A is invisible; visible violet leakage is separate. Fluorescence promptly re-emits absorbed energy as light, usually at a longer wavelength. A phosphor is a luminescent material, not the element phosphorus; the model uses a fluorescent phosphor. Some materials instead keep an afterglow (phosphorescence). Glow depends on both material and lamp wavelength. Ritter and Stokes supplied discoveries on the path to black lights; neither is presented as their sole inventor. The six illustrations are imagined museum scenes. Accept dictation and stop whenever you have had enough.

Materials: pencil, crayons if you like, and scrap paper. Print kid pages separately from this key.

Four clues to keep

The light we cannot see. · Quiz answer key

1. What kind of unseen light does an ordinary black light mostly give out?

UV-A: ultraviolet light near visible violet.

2. What does the star's fluorescent phosphor do?

It absorbs UV-A and emits visible light.

3. What is the name for this quick re-glow?

Fluorescence.

4. Why do some objects glow under a black light while others do not?

Suitable phosphors or fluorescent dyes can glow; other materials may show no noticeable fluorescence under that lamp.

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